Course Title: Case Studies in AI for Business 2026

Lecturer: Dr. Ahmed El-Seddawy

By : Mohamed El-Hawiet [23227076]

AI-Based Customer Segmentation and Targeted Marketing

Overview

I. Industry & Business Context
II. Problem Statement
III. AI Overview
IV. Analytical Applies Techniques
V. Digital Transformation Analysis
VI. ERP Integration (Odoo)
VII. Business Impact & Value Creation
VIII. Risks, Ethical Issues & limitations
IX. Recommendations & future Scope

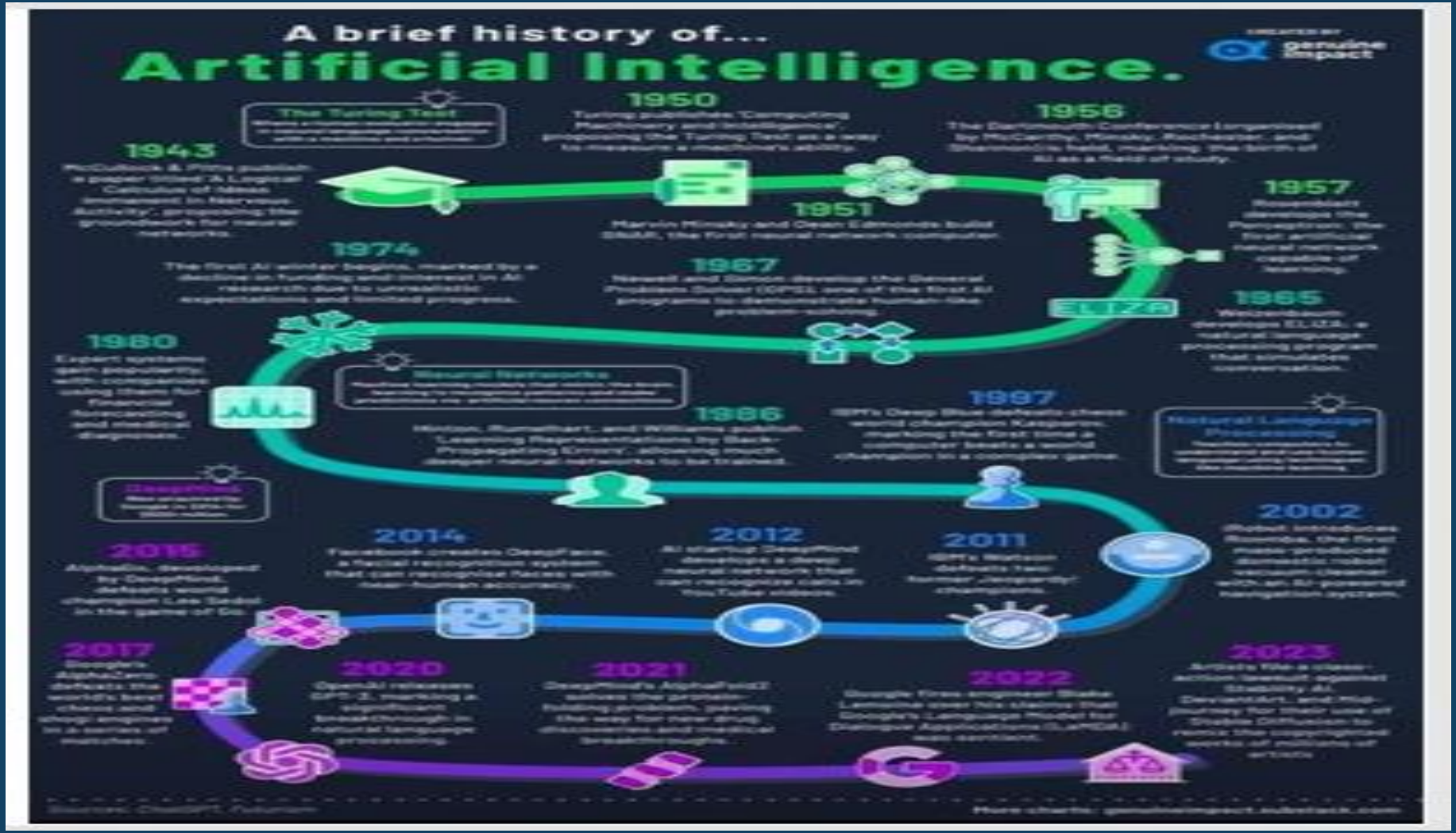
Industry & Business Context :

- The Egyptian Satellite Company, Nilesat was established in 1996 as the 1st Egyptian satellite operator in the MENA region, as it was also the 1st digital satellite owner in the MENA region & one of the first similar organizations in the whole world, Nilesat has a market share about 60% in the middle east & north Africa satellite & media market, broadcasting around 1200 services to reach around 160 million point of viewing all around the MENA region.
- Nilesat faced a very critical market decrease during 2021-2022 due to customer new behaviors –specially after covid19- which leaded to less reactions to Nilesat marketing campaigns, also the cost of gaining new customers increases which leaded in the end to loss some customers & decreasing the organization revenue, which is not accepted market position for a brand name like Nilesat.

Problem Statement:

- Nilesat is facing a huge competition from the national satellite operators, as Arabsat & Eshiel Sat beside the international operators, as Eutelsat & SES. Also competition is coming more from the digital platforms which is using the internet to target new customers throw the 2nd screens, as mobiles, tablets & laptops, Also the satellite companies suffer from very high cost for using the best technologies beside the dynamic market competitions.
- Nilesat needed a new strategy to adapt with the new market needs & to reach a better customer satisfaction. This paper will give some advices & work flow to implement the AI tools with the machine learning, digital transformation & smart forecasting beside using the best ways of reaching the benefits of using the organization resources to reach the targeted KPIs & customer segmentations based on the personal data.

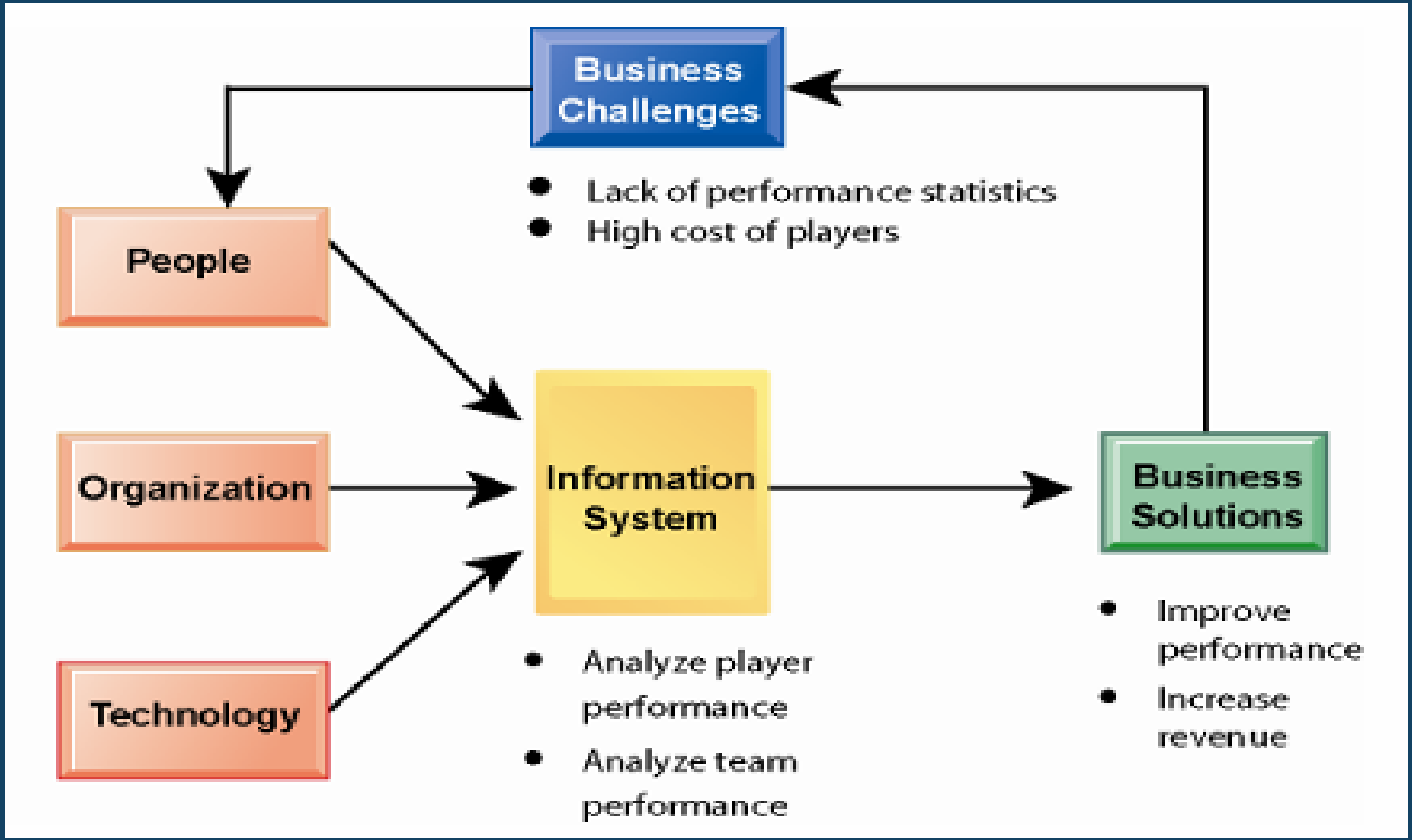
AI Overview:


A brief history of...
Artificial Intelligence.
genuine impact
The Turing Test
1943
1950
1951
1956
1957
1965
ELIZA
1967
1974
1980
1986
1997
2002
2011
2012
2014
2015
2017
2020
2021
2022
2023

AI Overview:

- The integration of the Artificial Intelligent (AI) in the satellite companies gives unlimited chances to increase the revenues & activate the customer's interactions. This paper studies the strategic applications that can use the artificial intelligent tools for customer segmentations & targeting new markets for the Egyptian satellite company, Nilesat.
- The Artificial Intelligent appears as a huge analytic tool in the organizations work flow & management throw big number of applications, which leads to a new understanding of the organizations to it's customers and direct interaction with them. In the Communication & media fields the customer analytics based on AI tools gives a very accurate market/customer segmentation, beside forecasting for the customer behaviors & giving customized services.

Analytical Applies Techniques:



Analytical Applies Techniques:

- Type of Data used:

Structured / Unstructured / Real-time Data:

- Data Sources:

ERP Systems / CRM & External Systems / IOT Data / Social Date

- Descriptive / Diagnostic / Predictive / Prescriptive Analytics:
- Key Performance Indicators (KPIs):

Customer Acquisition KPIs / Marketing & sales Effectiveness KPIs / Operational / Financial KPIs

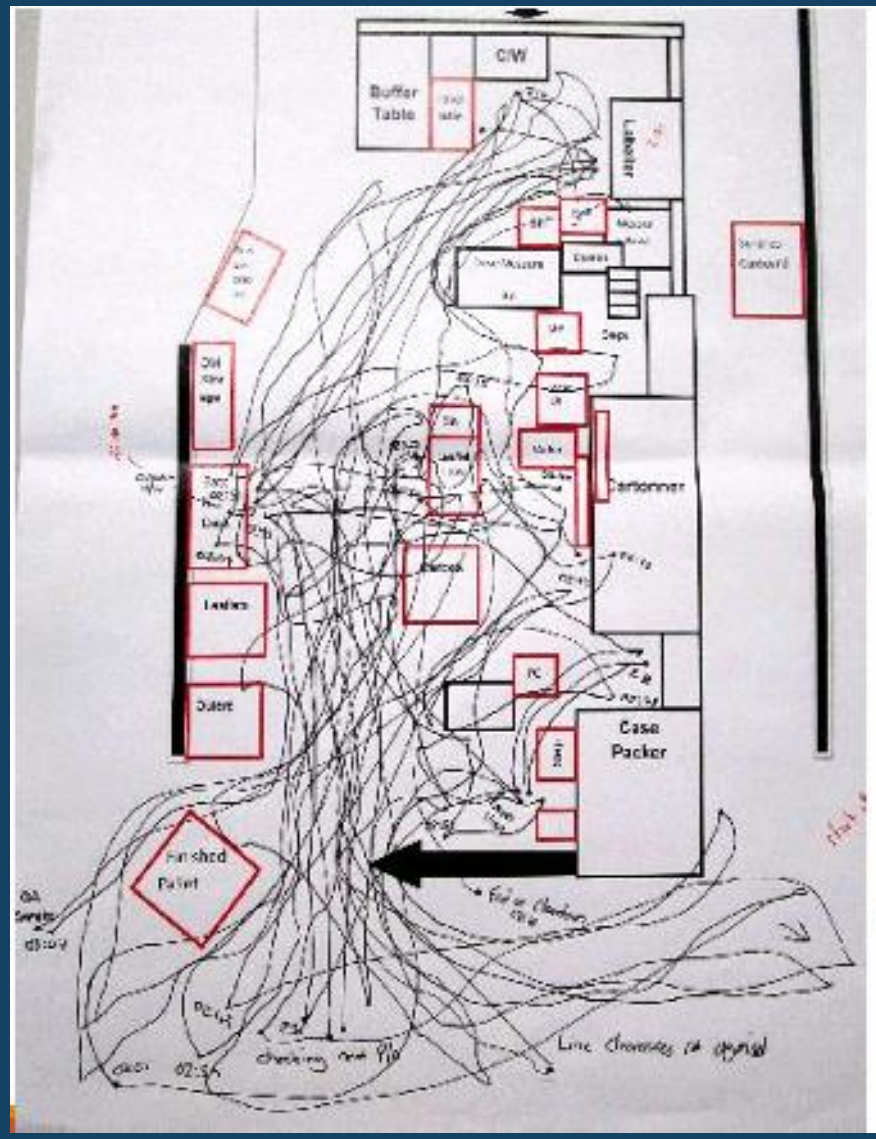
- Dashboards, Charts & Analytical models:

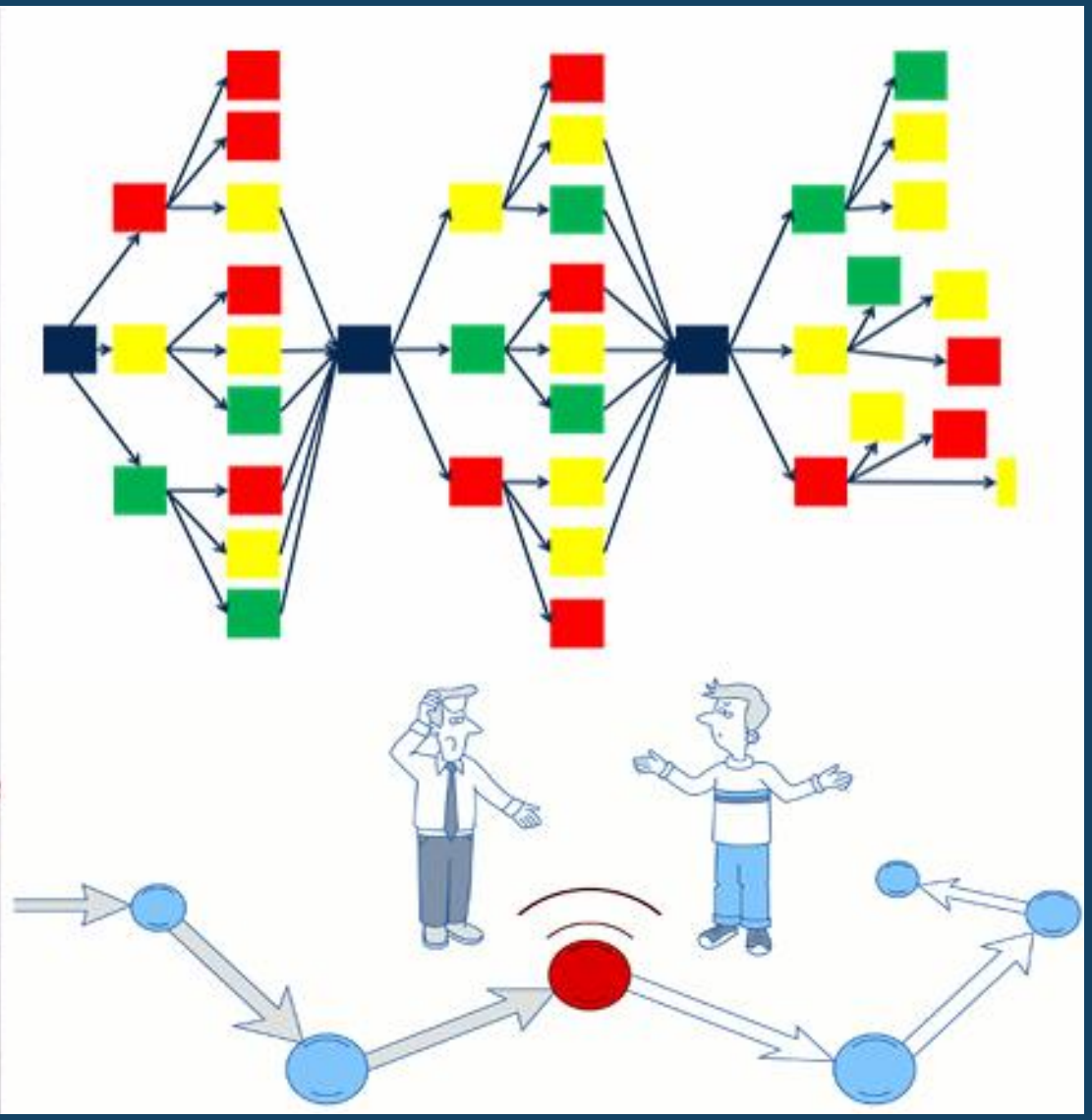
Executive Dashboard / Customer segmentation Model / Churn Prediction dashboard

- Decision making using data mining techniques:

Association Rule Mining / Classification & clustering / Time Series detection

Digital Transformation Analysis:





Digital Transformation Analysis:

- Changes in business process & workflows:
- Manual / Repetitive tasks automation:
- Impact on organizational culture & decision making:
- Customer experience improvements:
- Before & After digital transformation comparison:

FROM → Manual customer analysis ….. TO → AI customer segmentations

FROM → Limited market insights ….. TO → Real time analytics dashboards

FROM → Generic marketing campaigns ….. TO → Personalized targeted marketing

FROM → Classical slow decision making ….. TO → data driven strategic decisions

ERP Integration (Odoo):

ERP Integration (Odoo):

- ERP modules involved:

Nilesat uses Odoo as an ERP system with its several modules as Sales, Marketing, finance, CRM & data analytics which allows tracking contracts, billing invoices, marketing campaigns & related customer data, the integration of those models provides structured data that is used in AI analytics.

- Integration between AI & ERP systems:

AI tools uses ERP database information to analyze previous customer transaction datasets, campaign results & contracts details. Also machine learning models can utilize ERP data to produce insights about the potential marketing opportunities & the related customer behavior. This will support operational & strategic decisions.

ERP Integration (Odoo):

- Data flow between AI tools & ERP:

CRM models →Customer data → AI analytics platform → segmentation insights & marketing recommendations → ERP system → supports campaign planning & customer relationship management.

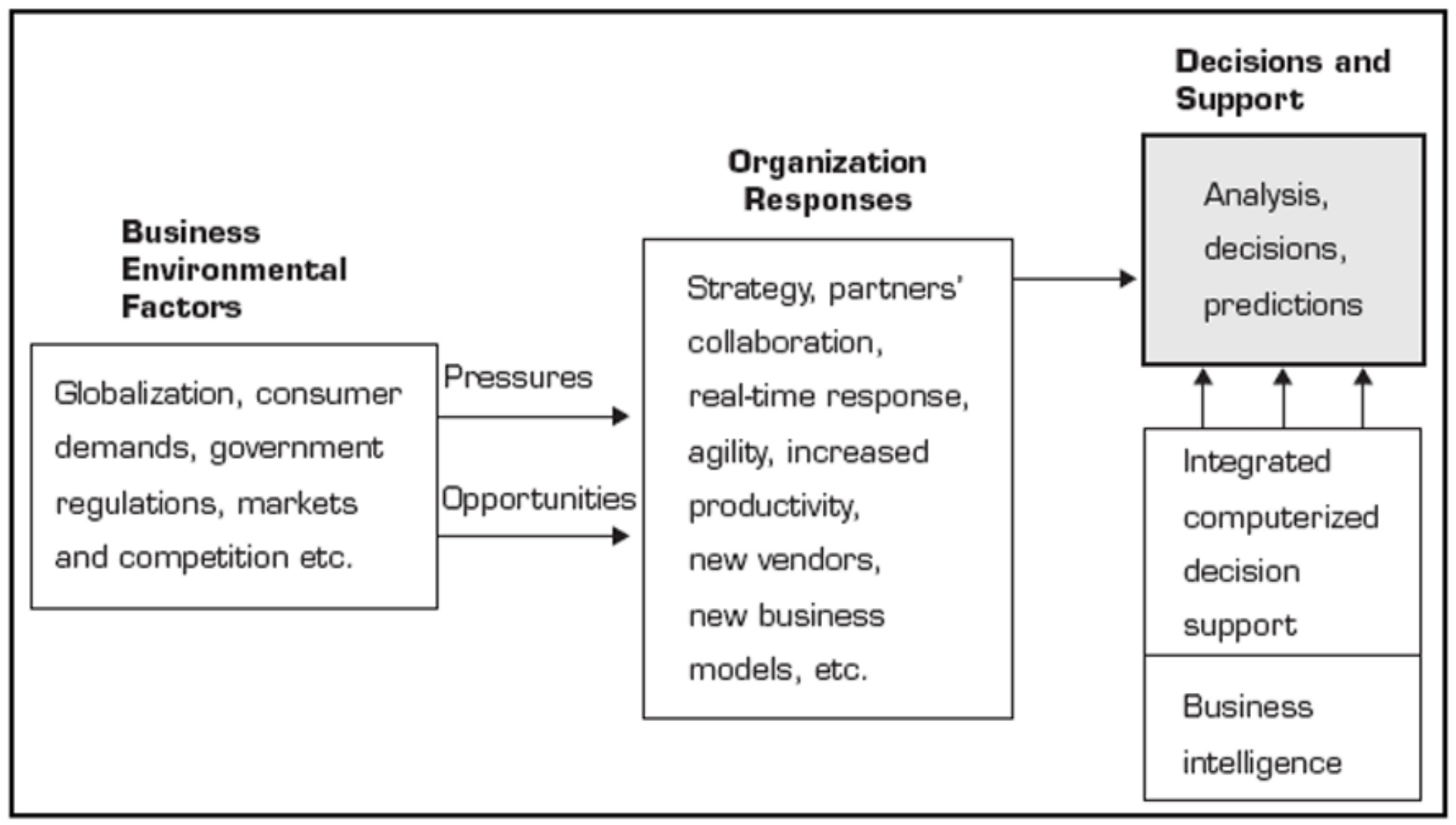
- Benefits of ERP & AI integration:

ERP+AI integration improves data accuracy, enhances decision making speed & leads to predictive marketing strategies, supports coordination between different departments which creates a unified digital ecosystem & improve operational efficiency.

- ERP implementation or customization challenges:

Any ERP system (as Odoo) needs customization to support AI tools, Data integration issues, organization culture, employee resistance & high implementation costs are the main reasons to slow down adoption process. Accurate AI insights need high data quality.

Business Impact & Value Creation:



Business Impact & Value Creation:

- Financial impact:

Using AI customer segmentation allows organizations to increase marketing efficiency & maximize revenue

- Operational efficiency improvements:

Automating the data analyzing process & creating marketing plans results in reducing the operational delays.

- Strategic advantages:

AI data driven services can be a competitive advantage for Nilesat customers. It will allow Nilesat to understand more it’s regional market demand & optimize satellite capacity usage.

- Scalability & long-term sustainability:

Using AI & ERP systems will help Nilesat to grow its business. Increasing accurate customer data leads to more valuable AI models will supports long-term digital transformations & sustainable competitive advantage in the satellite & media markets.

Risks, Ethical Issues & limitations:

Risks, Ethical Issues & limitations:

- Data privacy & security concerns:

The General Data protection regulation (GDPR).

- Bias AI models:

Non comprehensive used data may result in biased AI final outcome.

- Regulatory issues:

Nilesat should comply its data analysis to the telecom industry regulations (national & international), failing in this may result in legal consequences.

- Technology adoption risks:

Organization culture & employees normally resist the new technology.

- AI & ERRRP system limitations:

As poor data quality will lead to inaccurate predictions.

- ISO related controls:

Nilesat owns ISO 90001 certification & planning to gain ISO 27001.

Recommendations & future Scope:

- Strategic Recommendations:

Nilesat should invest more in the AI tools & analytical platforms, also it needs to give training for its teams to help them to upscale their skills.

- Advanced AI analytics & upgrades:

Nilesat systems should be upgraded to predict its customers needs using AI tools to understand more the MENA market. Those new technologies will give more opportunities in the market.

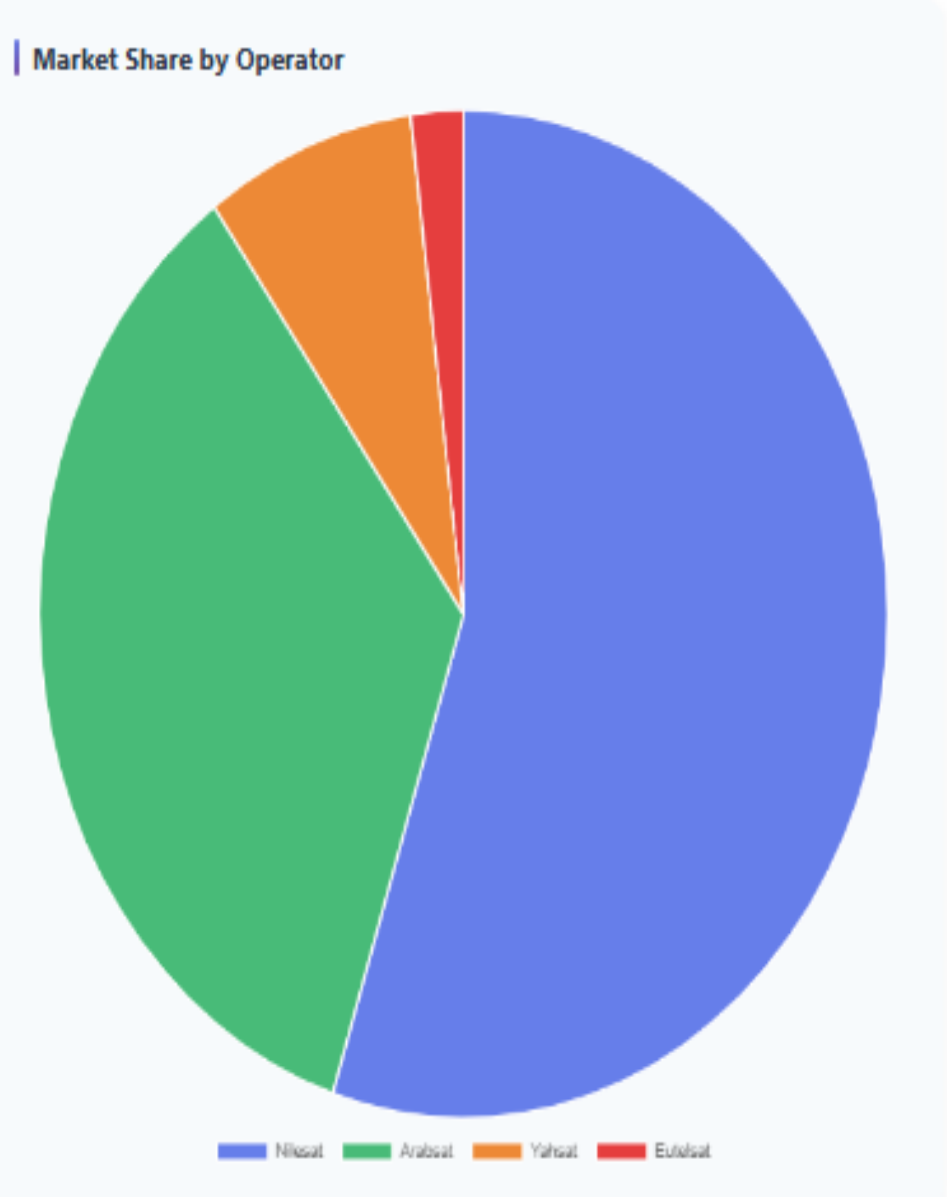
- Digital transformation future opportunities:

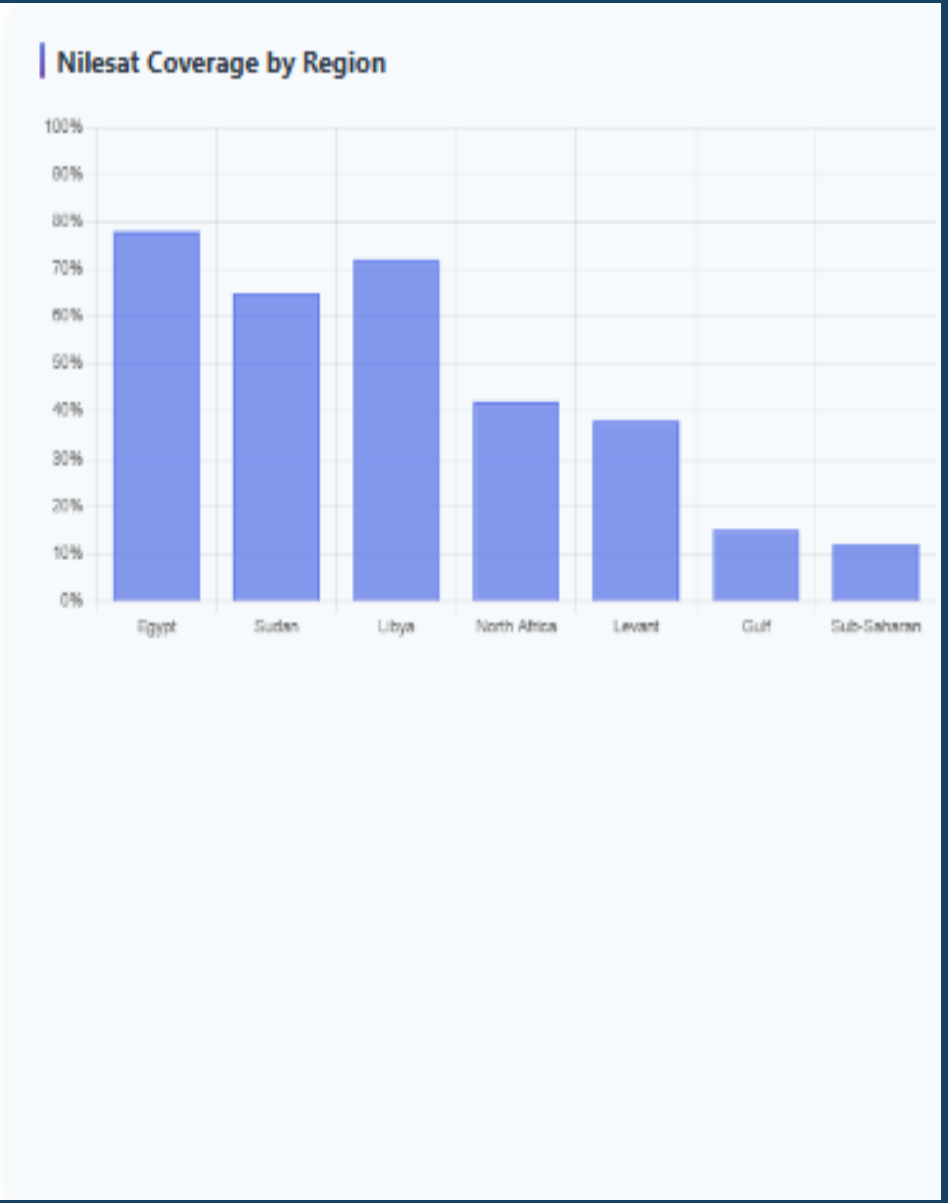
Combing AI with the cloud computing & big data platforms will help Nilesat to offer more digital services.

- AI / ERP capabilities expansion:

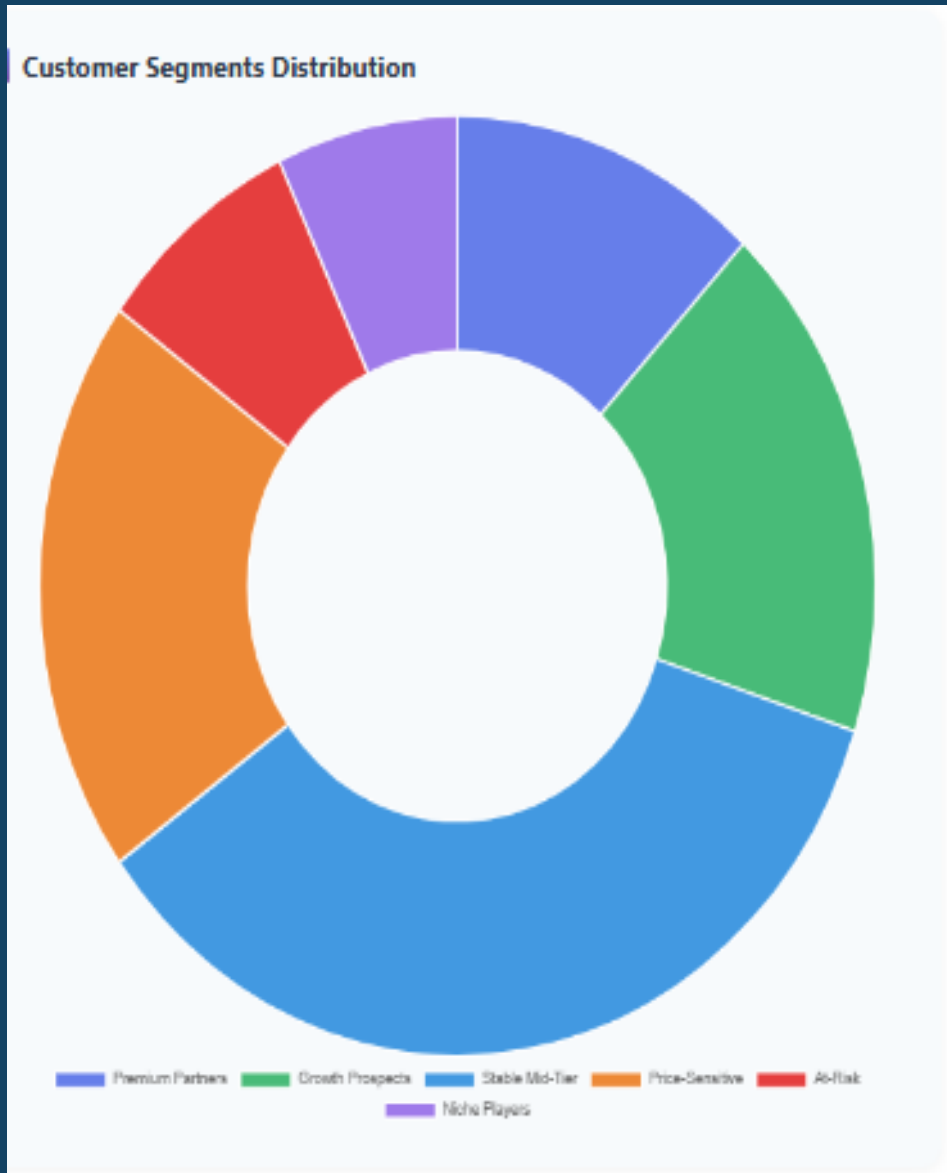
Nilesat should implement AI with its systems, Creating automatic reports using AI tools will make decisions more accurate. This integration will make Nilesat an intelligent enterprise, offering up-to-date services & maintain its leading position in the market.

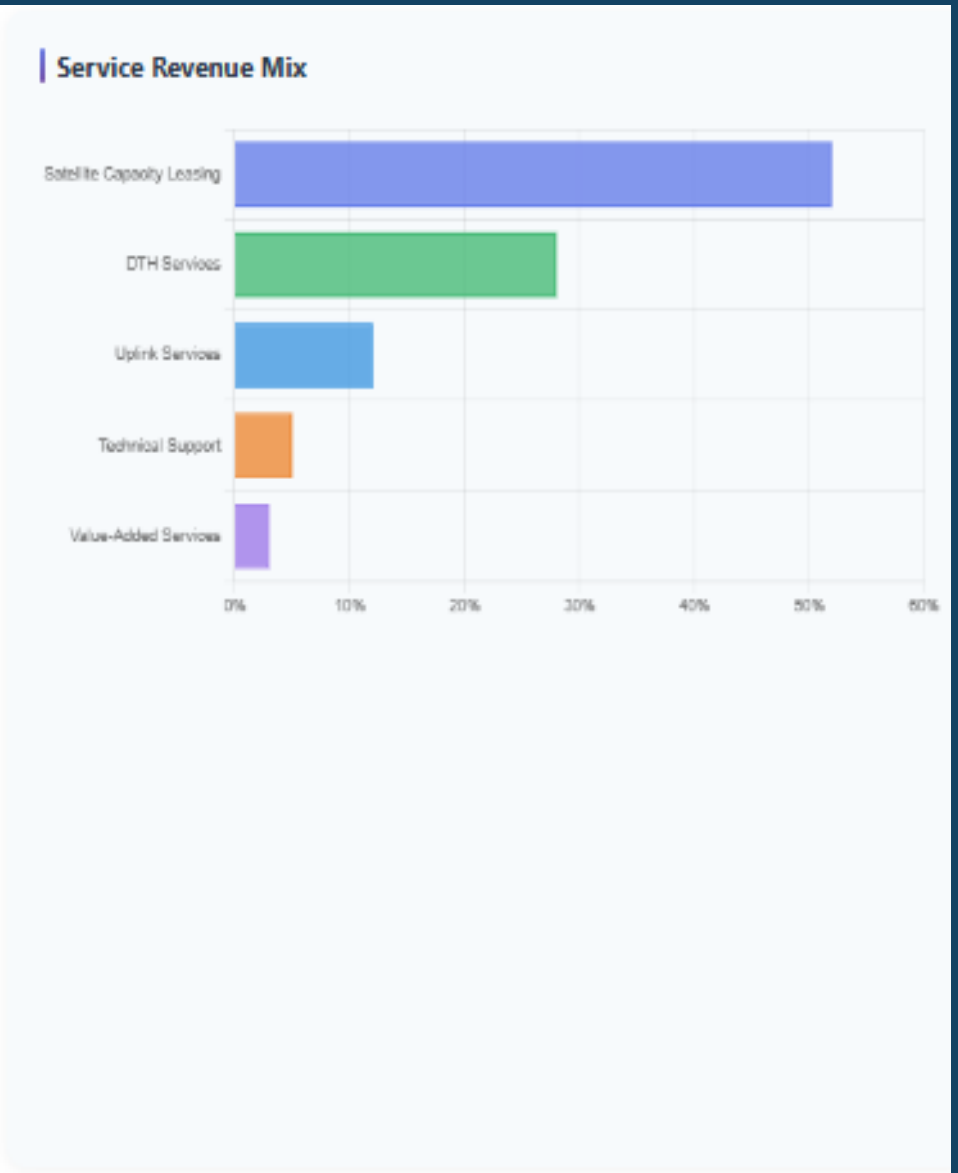
Sample of the Dash Boards :





Sample of the Dash Boards :





Sample of the Data Set:

Operator	Market Share (%)	Channels	Viewers (Millions)	Revenue 2024 ($M)	Revenue 2023 ($M)	Revenue 2022 ($M)	YoY Growth (%)	Regional Strength	Primary Target Markets
Nilesat	55%	1200	160	$215	$198	$185		North Africa & Levant	Egypt (78%), Libya (72%), Sudan (65%), North Africa (42%), Levant (38%)
Arabsat	35%	800	112	$142	$135	$128		Gulf States & Levant	Saudi Arabia (62%), UAE (55%), Kuwait (58%), Gulf (50-65%)
Yahsat	8%	150	26	$38	$35	$32		UAE & Sub-Saharan	UAE (30%), Sub-Saharan Africa (20%), Central Asia (15%)
Eutelsat	2%	250	6.4	$8	$9	$10		North Africa & Europe	Morocco (30%), Algeria (18%), Tunisia (15%), Europe (various)

Sample of the Data Set:

Country/Region	Market Priority	Penetration Rate (%)	Population (M)	TV Households (M)	Nilesat Subscribers (M)	Growth Potential	Revenue 2024 ($M)
Egypt	Primary	78%	104.00	24.00	18.72	Medium	$95
Sudan	Primary	65%	45.00	9.00	5.85	High	$22
Libya	Primary	72%	6.80	1.50	1.08	High	$12
Algeria	Secondary	42%	44.00	10.00	4.20	Very High	$18
Tunisia	Secondary	38%	12.00	2.80	1.06	High	$8
Morocco	Secondary	35%	37.00	8.50	2.98	Very High	$15
Jordan	Secondary	40%	10.00	2.20	0.88	Medium	$6
Lebanon	Secondary	42%	6.80	1.50	0.63	Medium	$5
Palestine	Secondary	35%	5.00	1.10	0.39	Medium	$3
Saudi Arabia	Emerging	15%	35.00	7.50	1.13	Low	$8
UAE	Emerging	12%	9.90	2.00	0.24	Low	$5
Sub-Saharan Africa	Emerging	12%	1100.00	220.00	26.40	Very High	$18

“If you can film an idea in your mind, follow that film idea shot for shot, scene for scene, that idea is worth making.

- Craig Mapp

Thank You